Market Outcome Model

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Key Finding

Macroeconomic variables do not reliably predict equity returns, but they do predict downside risk and volatility with meaningful out-of-sample accuracy. The strongest predictive power appears in drawdown and volatility classification, indicating that macro signals are more useful for risk management than for directional return forecasting.

1. Objective

The purpose of this project is to test whether macroeconomic regime information can be used to predict future market risk outcomes rather than only drive portfolio allocation decisions. Instead of mapping macro conditions directly to asset weights, this model estimates whether the current macro backdrop is associated with a higher probability of positive future returns, elevated drawdowns, or volatility spikes.

The target asset is SPY. The model evaluates whether macro regime variables such as inflation, growth, policy, yield curve conditions, and credit spreads contain useful information about forward 12-month equity returns and forward risk outcomes.

This project extends the earlier macro allocation model. The first model focused on portfolio construction. This one focuses on conditional market outcome prediction.

2. Forward Outcome Construction

Three forward-looking outcomes are constructed from monthly SPY returns. The first is the forward 12-month cumulative return. The second is the worst drawdown experienced over the next 12 months. The third is realized volatility over the next 6 months.

These outcomes allow the model to move from descriptive regime analysis to formal prediction of future market behavior.

Table 1: Preview of Forward Outcome Variables

Date	Regime	SPY_Ret	FwdRet_12M	MaxDD_12M	Vol_6M	Up12M	DD10_12M	VolSpike_6M
2004-01-31	Reflation	0.020	0.061	-0.032	0.075	1	0	0
2004-02-29	Goldilocks	0.014	0.069	-0.032	0.071	1	0	0
2004-03-31	Goldilocks	-0.013	0.063	-0.032	0.072	1	0	0
2004-04-30	Goldilocks	-0.019	0.064	-0.039	0.066	1	0	0
2004-05-31	Reflation	0.017	0.079	-0.039	0.086	1	0	0
2004-06-30	Reflation	0.019	0.061	-0.039	0.091	1	0	0
2004-07-31	Reflation	-0.032	0.139	-0.039	0.080	1	0	0
2004-08-31	Goldilocks	0.002	0.125	-0.039	0.078	1	0	0
2004-09-30	Goldilocks	0.010	0.123	-0.039	0.093	1	0	0
2004-10-31	Reflation	0.013	0.083	-0.039	0.102	1	0	0
2004-11-30	Reflation	0.045	0.082	-0.039	0.091	1	0	0
2004-12-31	Reflation	0.030	0.048	-0.037	0.080	1	0	0

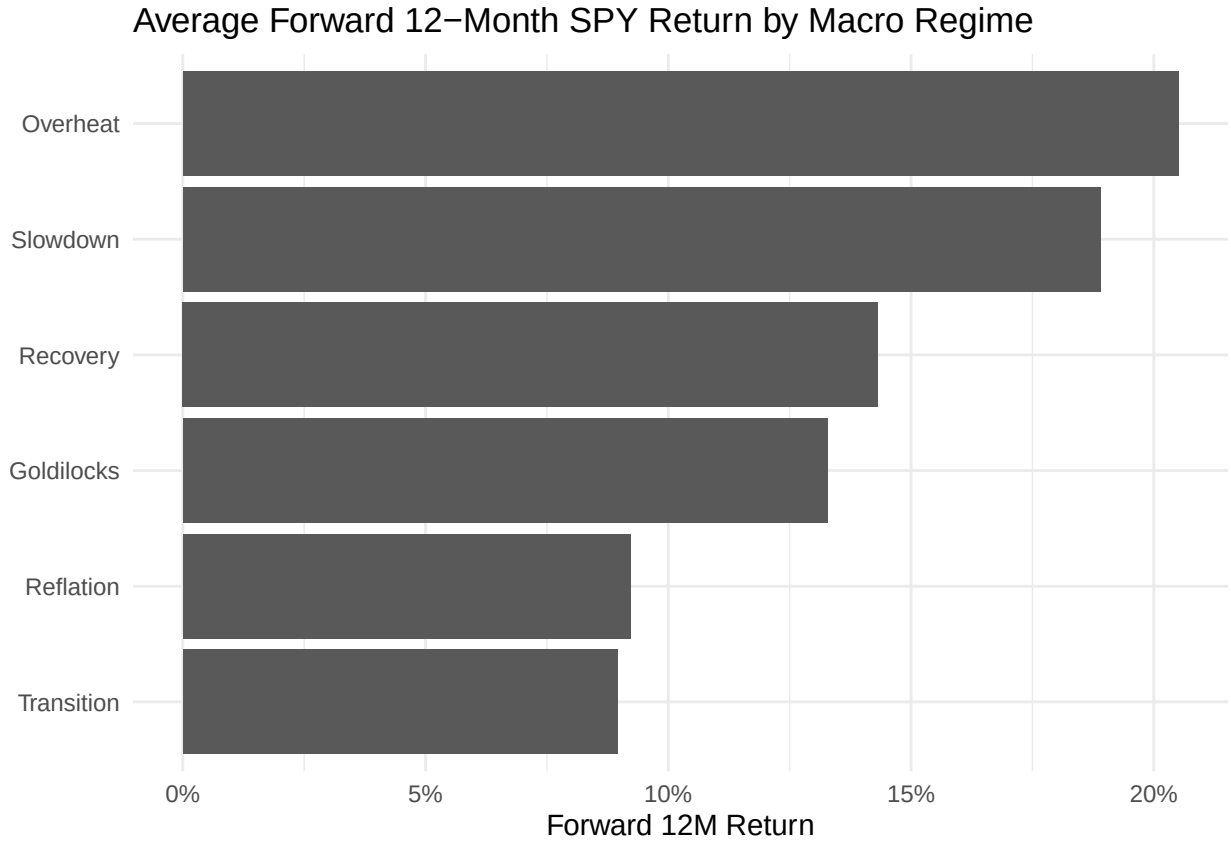
3. Regime-Level Outcome Analysis

Before fitting a predictive model, it is necessary to verify that macro regimes are associated with meaningfully different forward outcomes. If regimes do not differentiate future returns or risk, then predictive modeling is not justified.

Table 2: Forward SPY Outcomes by Macro Regime

Regime	N	AvgRet_12M	MedRet_12M	P_Up12M	P_DD10_12M	AvgMaxDD_12M	AvgVol_6M	P_VolSpike_6M
Transition	68	0.090	0.148	0.770	0.274	-0.098	0.121	0.172
Goldilocks	62	0.133	0.129	0.952	0.113	-0.061	0.111	0.194
Reflation	55	0.092	0.094	0.782	0.400	-0.086	0.128	0.273
Recovery	40	0.143	0.156	0.943	0.486	-0.105	0.131	0.256
Slowdown	21	0.189	0.214	0.810	0.381	-0.132	0.193	0.476
Overheat	7	0.205	0.180	1.000	0.000	-0.073	0.143	0.429
LateCycle	5	0.241	0.184	1.000	0.000	-0.048	0.107	0.000
SoftLanding	4	0.107	0.118	1.000	1.000	-0.127	0.136	0.250
Stagflation	4	-0.242	-0.231	0.000	1.000	-0.403	0.226	1.000

Regime-specific summaries for rare states should be interpreted cautiously because several regimes have very small sample counts.



Macro regimes exhibit materially different forward return and risk distributions. This validates the use of regime information as predictive input. The most important implication is not that regimes forecast returns precisely, but that they meaningfully separate benign and adverse future market conditions.

4. Modeling Dataset Construction

Table 3: Target Event Rates in Modeling Sample

Positive_12M_Return	Drawdown_10pct_12M	Volatility_Spike_6M
0.846	0.311	0.26

Table 4: Feature Correlation Matrix (Part 1)

	infl_score	grow_score	pol_score	curve_score	cred_score	macro_score	cpi_yoy
infl_score	1.00	0.01	-0.25	-0.08	0.20	-0.26	0.72
grow_score	0.01	1.00	-0.10	0.13	0.38	0.81	0.02
pol_score	-0.25	-0.10	1.00	-0.34	0.02	-0.40	-0.24
curve_score	-0.08	0.13	-0.34	1.00	-0.20	0.45	-0.21
cred_score	0.20	0.38	0.02	-0.20	1.00	0.41	0.04
macro_score	-0.26	0.81	-0.40	0.45	0.41	1.00	-0.23
cpi_yoy	0.72	0.02	-0.24	-0.21	0.04	-0.23	1.00
unrate_6m	-0.25	-0.54	-0.01	-0.02	-0.25	-0.32	-0.25
payems_3m	0.26	0.48	0.02	-0.04	0.24	0.25	0.28
real_ff	-0.47	-0.22	0.62	-0.47	0.07	-0.27	-0.58
term_10y3m	-0.12	0.10	-0.26	0.75	-0.09	0.38	-0.29
baa_aaa	-0.26	-0.56	0.04	0.11	-0.60	-0.42	-0.26
baa_10y	-0.33	-0.37	-0.19	0.40	-0.47	-0.07	-0.46

Table 5: Feature Correlation Matrix (Part 2)

	unrate_6m	payems_3m	real_ff	term_10y3m	baa_aaa	baa_10y
infl_score	-0.25	0.26	-0.47	-0.12	-0.26	-0.33
grow_score	-0.54	0.48	-0.22	0.10	-0.56	-0.37
pol_score	-0.01	0.02	0.62	-0.26	0.04	-0.19
curve_score	-0.02	-0.04	-0.47	0.75	0.11	0.40
cred_score	-0.25	0.24	0.07	-0.09	-0.60	-0.47
macro_score	-0.32	0.25	-0.27	0.38	-0.42	-0.07
cpi_yoy	-0.25	0.28	-0.58	-0.29	-0.26	-0.46
unrate_6m	1.00	-0.65	0.22	-0.03	0.39	0.34
payems_3m	-0.65	1.00	-0.20	-0.08	-0.37	-0.36
real_ff	0.22	-0.20	1.00	-0.42	0.06	-0.11
term_10y3m	-0.03	-0.08	-0.42	1.00	0.22	0.55
baa_aaa	0.39	-0.37	0.06	0.22	1.00	0.85
baa_10y	0.34	-0.36	-0.11	0.55	0.85	1.00

Feature abbreviation key

Short	Meaning
infl_score	inflation_score
grow_score	growth_score
pol_score	policy_score
curve_score	curve_score
cred_score	credit_score
macro_score	macro_score
cpi_yoy	cpi_yoy
unrate_6m	unrate_6m_delta_smooth
payems_3m	payems_3m_ann_smooth
real_ff	real_fedfunds
term_10y3m	term_spread_10y3m
baa_aaa	baa_aaa_spread
baa_10y	baa_10y_spread_smooth

The modeling dataset is structured so that each observation contains information available at a given point in time and the corresponding future realized market outcome. Several macro variables are correlated, so the model should be interpreted as predictive rather than causal.

5. Predictive Modeling Framework

The objective is to estimate the probability of future market outcomes using macroeconomic features available at each point in time.

A logistic regression framework is used to model binary outcomes. To avoid lookahead bias, an expanding window approach is applied. At each time step, the model is trained using only historical data and used to generate a prediction for the next observation.

6. Results

The model is evaluated using out-of-sample probability estimates. Accuracy provides a simple classification measure, while Brier score and log loss evaluate probability quality. AUC measures ranking ability, which is the most important metric in this setting.

Table 6: Out-of-Sample Prediction Performance

Target	observations	base_rate	avg_prob	accuracy	brier	logloss	auc
12M Drawdown <= -10%	134	0.261	0.353	0.716	0.210	0.815	0.734
6M Volatility Spike	134	0.306	0.265	0.754	0.196	0.945	0.729
Positive 12M Return	134	0.873	0.785	0.836	0.145	0.599	0.600

Table 7: Probability Separation Between Highest and Lowest Risk Buckets

Target	top_20	bottom_20	spread
12M Drawdown <= -10%	0.630	0.037	0.593
6M Volatility Spike	0.630	0.111	0.519
Positive 12M Return	0.852	0.741	0.111

Key Result

The model shows strong separation for adverse market events. For 12-month drawdowns, the top predicted-risk bucket experienced the event roughly 63% of the time versus 3.7% in the bottom bucket. For volatility spikes, the equivalent separation was roughly 63% versus 11%. By contrast, the return-direction model is materially weaker.

Return prediction performance is weak, with AUC near 0.60. This is consistent with efficient market behavior and reinforces that macro signals primarily capture risk regimes rather than directional alpha.

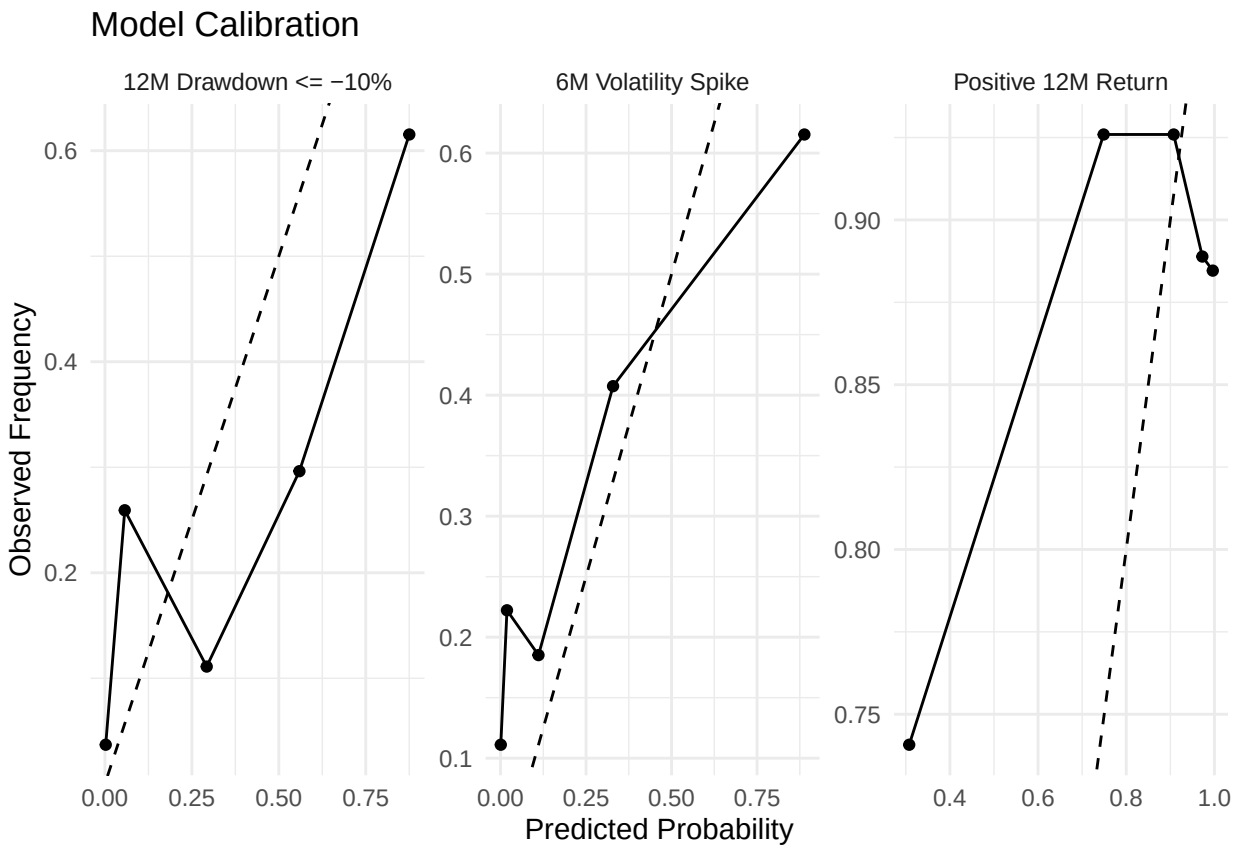
7. Probability Calibration

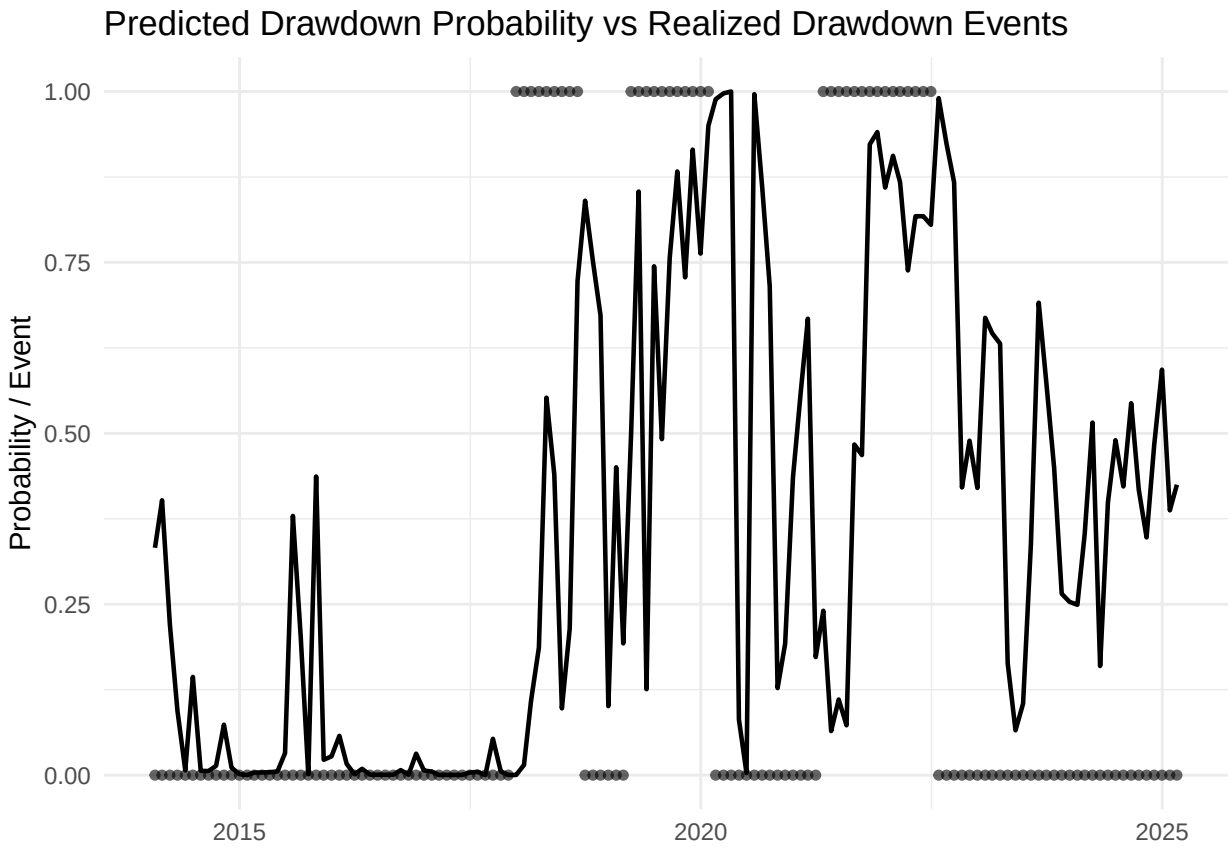
A useful probability model should assign higher predicted probabilities to events that occur more frequently. Calibration assesses whether predicted probabilities align with realized outcome frequencies.

Table 8: Calibration by Predicted Probability Quintile

Target	bucket	avg_prob	actual_rate	n
12M Drawdown <= -10%	1	0.002	0.037	27
12M Drawdown <= -10%	2	0.057	0.259	27
12M Drawdown <= -10%	3	0.293	0.111	27
12M Drawdown <= -10%	4	0.559	0.296	27
12M Drawdown <= -10%	5	0.875	0.615	26
Positive 12M Return	1	0.308	0.741	27
Positive 12M Return	2	0.749	0.926	27

Positive 12M Return	3	0.908	0.926	27
Positive 12M Return	4	0.973	0.889	27
Positive 12M Return	5	0.996	0.885	26
6M Volatility Spike	1	0.001	0.111	27
6M Volatility Spike	2	0.019	0.222	27
6M Volatility Spike	3	0.111	0.185	27
6M Volatility Spike	4	0.329	0.407	27
6M Volatility Spike	5	0.888	0.615	26





The calibration results show that the model's highest predicted-risk states generally correspond to higher observed event frequencies. This is especially important for drawdowns and volatility, where probability ranking matters more than exact classification.

8. Model Interpretation

To understand which macroeconomic variables drive predictions, the model is estimated on the full sample and coefficient estimates are examined.

Coefficient estimates indicate that growth, credit spreads, real policy rates, and term-structure variables are the most influential predictors of adverse market states in this specification.

9. Practical Use

This framework is most useful as a risk-classification tool. In practice, it could be used to reduce equity exposure, tighten risk limits, or increase defensive positioning during periods of elevated predicted drawdown or volatility risk.

10. Conclusion

This project demonstrates that macroeconomic regime information contains meaningful predictive power for future market risk outcomes. While return direction is difficult to forecast, drawdowns and volatility exhibit strong out-of-sample relationships with macro conditions.

The strongest results appear in drawdown and volatility classification, where both AUC and separation metrics indicate economically meaningful signal. These results suggest that macro-driven models are more effective for exposure control and risk management than for return forecasting.

Table 9: Full-Sample Logistic Regression Coefficients by Target

term	estimate	std_error	z_value	p_value	Target
(Intercept)	2.614	0.349	7.485	0.000	Positive 12M Return
inflation_score	-0.660	0.337	-1.959	0.050	Positive 12M Return
growth_score	1.008	0.346	2.910	0.004	Positive 12M Return
policy_score	0.331	0.315	1.051	0.293	Positive 12M Return
curve_score	-0.158	0.384	-0.411	0.681	Positive 12M Return
credit_score	0.864	0.351	2.461	0.014	Positive 12M Return
cpi_yoy	-0.501	0.695	-0.721	0.471	Positive 12M Return
unrate_6m_delta_smooth	0.591	0.387	1.526	0.127	Positive 12M Return
payems_3m_ann_smooth	-0.897	0.534	-1.678	0.093	Positive 12M Return
real_fedfunds	-0.861	0.635	-1.356	0.175	Positive 12M Return
term_spread_10y3m	-1.050	0.572	-1.837	0.066	Positive 12M Return
baa_aaa_spread	0.362	0.869	0.417	0.677	Positive 12M Return
baa_10y_spread_smooth	1.239	1.101	1.125	0.261	Positive 12M Return
(Intercept)	-1.052	0.173	-6.090	0.000	12M Drawdown <= -10%
inflation_score	0.352	0.251	1.403	0.161	12M Drawdown <= -10%
growth_score	-1.537	0.293	-5.254	0.000	12M Drawdown <= -10%
policy_score	-0.691	0.251	-2.747	0.006	12M Drawdown <= -10%
curve_score	-0.198	0.297	-0.665	0.506	12M Drawdown <= -10%
credit_score	0.218	0.241	0.903	0.366	12M Drawdown <= -10%
cpi_yoy	-1.126	0.530	-2.123	0.034	12M Drawdown <= -10%
unrate_6m_delta_smooth	-0.293	0.224	-1.307	0.191	12M Drawdown <= -10%
payems_3m_ann_smooth	0.196	0.236	0.831	0.406	12M Drawdown <= -10%
real_fedfunds	-0.912	0.475	-1.922	0.055	12M Drawdown <= -10%
term_spread_10y3m	1.194	0.337	3.549	0.000	12M Drawdown <= -10%
baa_aaa_spread	2.139	0.633	3.377	0.001	12M Drawdown <= -10%
baa_10y_spread_smooth	-3.282	0.810	-4.053	0.000	12M Drawdown <= -10%
(Intercept)	-1.357	0.193	-7.039	0.000	6M Volatility Spike
inflation_score	0.179	0.284	0.631	0.528	6M Volatility Spike
growth_score	-0.870	0.283	-3.080	0.002	6M Volatility Spike
policy_score	-0.309	0.274	-1.128	0.260	6M Volatility Spike
curve_score	-0.632	0.381	-1.660	0.097	6M Volatility Spike
credit_score	0.007	0.255	0.027	0.978	6M Volatility Spike
cpi_yoy	-1.970	0.648	-3.042	0.002	6M Volatility Spike
unrate_6m_delta_smooth	0.334	0.254	1.315	0.189	6M Volatility Spike
payems_3m_ann_smooth	-0.189	0.275	-0.688	0.492	6M Volatility Spike
real_fedfunds	-2.882	0.631	-4.567	0.000	6M Volatility Spike
term_spread_10y3m	0.435	0.358	1.215	0.224	6M Volatility Spike
baa_aaa_spread	2.732	0.722	3.783	0.000	6M Volatility Spike
baa_10y_spread_smooth	-3.690	0.903	-4.088	0.000	6M Volatility Spike

Future extensions could include alternative model specifications, additional features, regularization methods, or cross-asset applications.